A Three-Tier Knowledge Graph RAG for Vietnamese Legal Question Answering

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Abstract. Answering legal questions accurately requires navigating hierarchical document structures, following cross-references, and performing multi-hop reasoning—capabilities that standard Retrieval Augmented Generation (RAG) systems lack. This gap is especially pronounced in Vietnamese legal text, where tonal language, domain-specific abbreviations, and a rigid structural hierarchy (Law→Chapter→Article→Clause→Point) compound retrieval difficulty. We propose a three-tier knowledge graph RAG architecture that extends the Legal-Onto Rela-model with a formal knowledge model $\mathcal{M} = (\mathcal{O}, \mathcal{G}, \mathcal{T}, \Phi)$, integrating an OWL ontology for terminological knowledge, a domain-specific knowledge graph for assertional knowledge, and a document tree forest for structural navigation. The three retrieval tiers—LLM-guided tree traversal, dual-level semantic retrieval with intent-aware Personalized PageRank, and Reciprocal Rank Fusion—operate over these complementary representations. Experiments on 375 real Vietnamese legal questions show that our system achieves 75.2% Hit@10 (MRR 0.545), outperforming BM25 (26.7%), semantic search (15.8%), LightRAG (33.9%), and PageIndex (35.2%) by substantial margins. Ablation studies confirm each tier's additive contribution, with Tier 3 fusion providing 4.0 percentage points over Tier 1+2 alone. To our knowledge, this is the first system to combine ontology-enhanced knowledge graphs with multi-tier retrieval for Vietnamese legal question answering.

Keywords. Knowledge Graph, Legal Question Answering, Retrieval Augmented Generation, Ontology, Vietnamese NLP, Multi-Tier Retrieval

1. Introduction

Legal question answering (QA) systems serve a growing need as citizens and businesses seek to understand their rights and obligations. However, building effective systems for this domain presents challenges that distinguish it from general-purpose QA.

Legal documents exhibit **complex hierarchical structures**. A single law may organize content into parts, chapters, sections, articles, clauses, and points, where each level carries distinct legal scope. Understanding these structural relationships is essential for accurate retrieval: an article on shareholder voting rights only applies within the chapter governing joint-stock companies.

Legal texts also contain pervasive **cross-references**. An article defining establishment conditions may reference another specifying required documents, which in turn points to procedures defined elsewhere. Answering a practical question like "Can a foreign investor establish a limited liability company in Vietnam?" requires following reference chains across multiple articles to assemble conditions, procedures, and exceptions.

The **Vietnamese legal domain** introduces additional complexity. Vietnamese is a tonal language with compound words whose boundaries are ambiguous without segmentation. Legal practitioners use extensive abbreviations (*HĐQT* for "Họi dòng quan trị", *CTCP* for "Công ty co phàn") that general-purpose models do not recognize. Moreover, Vietnamese law follows a rigid hierarchy—Constitution, Codes, Decrees, Circulars—where document type determines legal authority and interpretation precedence.

Traditional RAG systems [2] rely on vector similarity between query and document embeddings [22]. While effective for many domains [15, 16], this approach faces three fundamental limitations for legal QA: (1) a **semantic gap** where legal terminology has precise meanings that general embeddings miss, (2) **no structural awareness** since vector similarity treats documents as flat text, and (3) **no reasoning capability** since dense retrieval cannot follow reference chains or apply legal inference.

No existing system addresses all three limitations simultaneously. Graph-based approaches like LightRAG [5] and GraphRAG [4] model entity relationships but ignore document hierarchy. PageIndex [14] navigates hierarchical structure but lacks semantic

reasoning. Legal ontology systems [6, 8] represent domain knowledge but are not integrated with modern retrieval architectures. The intersection of ontology-enhanced knowledge graphs, multi-tier retrieval, and Vietnamese legal processing remains unexplored.

We propose a **three-tier knowledge graph RAG** architecture that addresses these gaps through:

1. **LLM-guided tree traversal** through document hierarchies for structural navigation (Tier 1),
2. **Dual-level semantic retrieval** with intent-aware Personalized PageRank on a domain knowledge graph (Tier 2),
3. **Reciprocal Rank Fusion** for cross-tier merging with knowledge graph expansion (Tier 3).

Our contributions are as follows. First, we propose a three-tier RAG architecture that combines structural, semantic, and graph-based retrieval. Second, we construct a Vietnamese legal knowledge graph with 1,299 entities across 11 types and 2,577 relations across 30 types, with LLM-based extraction and Vietnamese slug normalization for deduplication. Third, we introduce a multi-intent Personalized PageRank algorithm that adapts edge weights based on detected query intents. Fourth, we conduct comprehensive experiments on 375 real legal questions demonstrating 75.2% Hit@10 (+41.3pp over the best baseline), with ablation studies confirming each tier's contribution.

The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 describes our methodology. Section 4 presents experiments. Section 5 discusses findings. Section 6 concludes.

2. Related Work

2.1. Retrieval-Augmented Generation

Retrieval-Augmented Generation (RAG) [2] enhances language models by grounding generation in retrieved evidence. The field has evolved from basic retrieve-then-read pipelines [3] to advanced architectures incorporating self-reflection [23], corrective retrieval [24], and hypothetical document generation [25]. Comprehensive surveys [15, 16] document this progression toward modular and agentic RAG systems. However, most advances target general-domain QA and do not address the structural and semantic requirements of legal text.

2.2. Knowledge Graphs in Information Retrieval

Knowledge graph (KG) augmented retrieval has gained traction through systems like GraphRAG [4], which constructs KGs from documents for multi-hop reasoning and global summarization, and LightRAG [5], which provides lightweight entity-relation extraction. KG embedding methods including TransE [19] and RotatE [30] learn entity representations in multi-relational graphs, while Personalized PageRank [12, 20] computes query-specific node importance. PageIndex [14] introduced LLM-guided tree traversal for hierarchical navigation using document structure summaries. These approaches address individual aspects—graph reasoning, structural navigation—but none combines them into a unified multi-tier framework.

2.3. Legal NLP and QA Systems

Legal information retrieval has progressed from lexical methods [1, 11] to neural approaches using domain-adapted models like LEGAL-BERT [27]. Ontology-based representations [8] formalize legal concepts and relations, with Legal-Onto [6] integrating the Rela-model [10] with conceptual graphs for Vietnamese Land Law. For Vietnamese NLP, PhoBERT [17] and VnCoreNLP [18] provide foundational capabilities, while Ngo et al. [7] designed a QA system using BM25, BERT [13], and legal domain knowledge. Broader surveys [28, 31] document the challenges of applying NLP to legal text, including specialized vocabulary and complex reasoning requirements. However, no existing system combines ontology-enhanced KGs with multi-tier retrieval for Vietnamese legal QA.

2.4. Positioning

Table 1 compares key features across systems. Our approach is the first to unify knowledge graph reasoning, domain ontology, tree navigation, intent-aware retrieval, and rank fusion in a single architecture.

Table 1: Feature comparison of RAG systems.

Feature	BM25	Light	Graph	Page	Ours
Knowledge Graph	–	✓	✓	–	✓
Multi-tier	–	–	–	–	✓
Intent-aware	–	–	–	–	✓
Domain Ontology	–	–	–	–	✓
Tree Navigation	–	–	–	✓	✓
Rank Fusion	–	–	–	–	✓

3. Methodology

3.1. Problem Formulation

Given a legal corpus $\mathcal{D} = \{d_1, d_2, \ldots, d_m\}$ where each document d_i contains articles $\{a_{i,1}, \ldots, a_{i,n_i}\}$ organized hierarchically, and a natural language query q, the task is to retrieve a ranked list of articles $A^* = [a_1^*, a_2^*, \ldots, a_k^*]$ that are relevant to answering q.

We construct a knowledge model $\mathcal{M}$ from $\mathcal{D}$ and

define retrieval as:

$$A^* = \Phi(q, \mathcal{M}) \quad \text{where} \quad \mathcal{M} = (\mathcal{O}, \mathcal{G}, \mathcal{T}, \Phi) \quad (1)$$

The model $\mathcal{M}$ integrates four components: ontology $\mathcal{O}$ for terminological knowledge, knowledge graph $\mathcal{G}$ for assertional knowledge, document forest $\mathcal{T}$ for structural knowledge, and retrieval functions Φ that operate over these representations.

3.2. System Overview

Figure 1 illustrates the architecture. The **offline phase** constructs $\mathcal{M}$ from legal documents through parsing, entity-relation extraction, and ontology generation. The **online phase** processes queries through ontology-based expansion (Stage 0) followed by three retrieval tiers that leverage $\mathcal{T}$, $\mathcal{G}$, and $\mathcal{O}$ respectively.

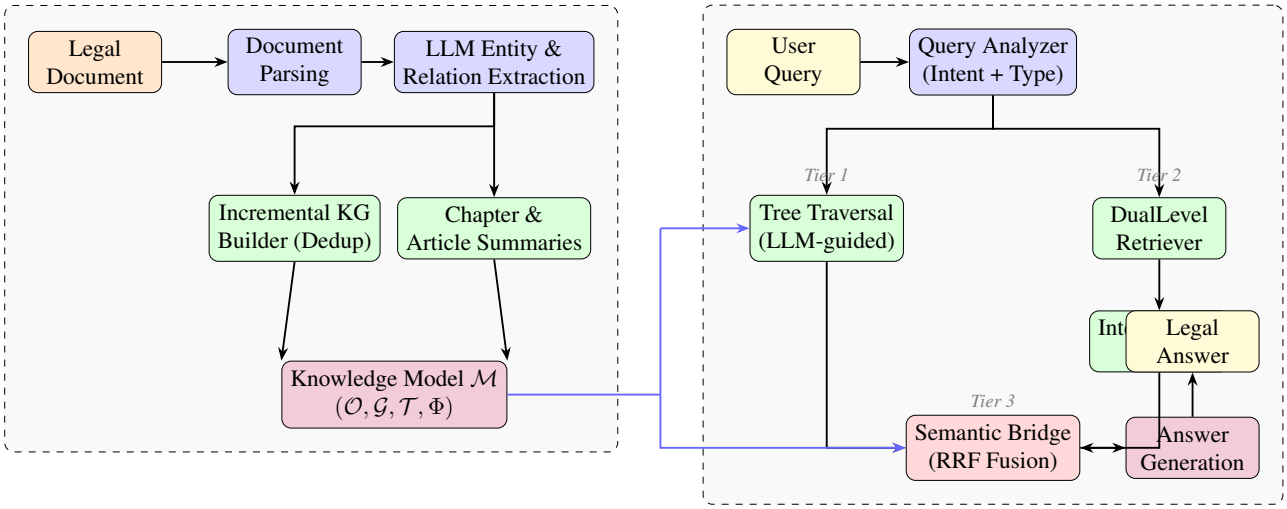

Figure 1: System architecture. The offline phase constructs knowledge model $\mathcal{M}$ from legal documents. The online phase processes queries through three tiers: dual-level semantic retrieval with PPR (Tier 2, runs first for global coverage), LLM-guided tree traversal (Tier 1), and RRF-based fusion with KG expansion (Tier 3).

3.3. Knowledge Representation Model

Our knowledge model extends the Legal-Onto framework [6], which integrates the Rela-model [10] with conceptual graphs for representing legal knowledge. Legal-Onto defines a knowledge structure $K = (C, R, \mathit{RULES}) + (\mathit{Key}, \mathit{Rel}, \mathit{weight})$, where C is a set of concepts, R is relations, *RULES* is inference rules, and $(\mathit{Key}, \mathit{Rel}, \mathit{weight})$ is a conceptual graph of key phrases. We extend this formulation by separating terminological knowledge from assertional knowledge, adding structural representation, and replacing manual inference rules with LLM-based retrieval.

Definition 1 (Legal Knowledge Model)**.** *Given a legal corpus $\mathcal{D}$, the knowledge model is:*

$$\mathcal{M} = (\mathcal{O}, \mathcal{G}, \mathcal{T}, \Phi) \quad (2)$$

where $\mathcal{O}$ is a domain ontology, $\mathcal{G}$ is a knowledge graph, $\mathcal{T}$ is a document forest, and Φ is a set of retrieval functions.

Ontology $\mathcal{O} = (C, P, H, L)$ represents terminological knowledge:

- $C = \{c_1, \ldots, c_n\}$: OWL classes representing legal entity types (e.g., `JointStockCompany`, `GovernmentAgency`)
- $P = P_{\text{obj}} \cup P_{\text{data}}$: Object and datatype properties (e.g., *governedBy*, *charterCapital*)
- $H \subseteq C \times C$: Class hierarchy via `rdfs:subClassOf`, forming a 4-level taxonomy
- $L : C \cup P \to$ Vietnamese: Label mapping from classes and properties to Vietnamese names (e.g., `JointStockCompany` $\mapsto$ "công ty co phàn")

This separates the terminological layer (TBox) from Legal-Onto's flat concept set C, enabling schema-level reasoning such as class hierarchy traversal for query expansion.

Knowledge Graph $\mathcal{G} = (E, R)$ represents assertional knowledge:

- $E = \{e \mid e = (\mathit{id}, \mathit{name}, \mathit{type}, \mathit{conf}, \mathit{src})\}$: Entities with slug-normalized IDs, 11 typed categories, confidence scores, and source provenance. Entity types include Organization, Person/Role, Legal Term, Legal Reference, Monetary, Percentage, Duration, Condition, Action, and Penalty.

- $R = \{r \mid r = (e_i, \textit{pred}, e_j, \textit{evid}, \textit{conf})\}$: Typed relations with evidence text and confidence. The 30 relation types carry formal properties:
  - Symmetric: $\{\textit{related_to}, \textit{contradicts}, \textit{synonym}, \textit{excludes}\}$
  - Transitive: $\{\textit{is_a}, \textit{part_of}, \textit{contains}, \textit{precedes}\}$

Compared to Legal-Onto's conceptual graph $(\textit{Key}, \textit{Rel}, \textit{weight})$ which stores key phrases with binary similarity vectors, $\mathcal{G}$ provides richer entity typing, confidence tracking, and evidence-backed relations grounded in specific articles.

Document Forest $\mathcal{T} = \{T_1, \ldots, T_m\} \cup X$ represents structural knowledge:

- Each T_i is a rooted tree for document d_i with nodes at six levels: DOCUMENT $\rightarrow$ CHAPTER $\rightarrow$ SECTION $\rightarrow$ ARTICLE $\rightarrow$ CLAUSE $\rightarrow$ POINT
- $X \subseteq \{(n_a, n_b, \textit{conf}) \mid n_a \in T_i, n_b \in T_j, i \neq j\}$: Cross-reference edges between documents

The document forest is absent from Legal-Onto, which does not model hierarchical navigation. This component enables Tier 1's structural traversal.

Retrieval Functions $\Phi = \{\phi_0, \phi_1, \phi_2, \phi_3\}$ replace Legal-Onto's manual inference rules (*RULES*):

$$\begin{aligned}
&\phi_0 : Q \times \mathcal{O} \rightarrow Q' && \text{(ontology expansion)}\\
&\phi_1 : Q' \times \mathcal{T} \rightarrow 2^A && \text{(tree traversal)}\\
&\phi_2 : Q' \times \mathcal{G} \times \mathcal{O} \rightarrow 2^A && \text{(semantic retrieval)}\\
&\phi_3 : 2^A \times 2^A \times \mathcal{G} \rightarrow A^* && \text{(fusion)}
\end{aligned} \tag{3}$$

Where Legal-Onto applies hand-crafted deductive rules (e.g., "if $[\Theta$ is symmetric$]$ and $[k_1 \Theta k_2]$ then $[k_2 \Theta k_1]$"), our retrieval functions leverage LLM reasoning for flexible multi-hop inference while preserving formal relation properties (symmetry, transitivity) in the graph structure.

3.4. Knowledge Graph Construction

The offline phase builds $\mathcal{G}$ through a semi-automated pipeline:

Step 1: Document Parsing. Legal documents are parsed into hierarchical trees following Vietnamese legal formatting standards (Article 11, Circular 25/2011/TT-BTP). Each structural element is stored with position, parent-child relationships, cross-references, and full text, forming the document forest $\mathcal{T}$.

Step 2: Unified Entity-Relation Extraction. A single LLM call per article extracts both entities and relations simultaneously. The prompt specifies 11 entity types and instructs the model to produce evidence-grounded relations with source text citations.

Step 3: Incremental KG Building. Extracted entities are deduplicated using Vietnamese slug normalization: Unicode NFD decomposition, diacritics removal, and lowercasing (e.g., "Công ty co phàn" $\rightarrow$ *cong-ty-co-phan*). Entity resolution uses a 4-pass pipeline: exact cache lookup, synonym matching (17 entries), abbreviation expansion (25 entries), and new slug creation. Confidence scores are tracked across multiple extraction occurrences.

Step 4: Summary Generation. LLM-generated summaries for each chapter and article enable tree traversal (Tier 1) to make informed navigation decisions without reading full article text.

3.4.1. Intent-Aware Edge Weights

Different query intents prioritize different relation types during graph traversal. Table 2 defines intent-specific weight multipliers for the PPR algorithm.

Table 2: Intent-relation weight matrix.

Relation	Def.	Proc.	Right	Obl.	Pen.
defines_as	1.0	0.5	0.5	0.5	0.5
includes	0.95	0.5	0.5	0.5	0.5
requires	0.5	1.0	0.5	0.85	0.5
has_authority	0.5	0.5	1.0	0.5	0.5
has_obligation	0.5	0.5	0.5	1.0	0.5
has_penalty	0.5	0.5	0.5	0.5	1.0

For definition queries, *defines_as* and *includes* receive weight 1.0 and 0.95 respectively. For procedure queries, *requires* is prioritized. This allows PPR to traverse paths most relevant to query intent.

3.5. Three-Tier Retrieval

3.5.1. Stage 0: Query Analysis

Before retrieval, the query analyzer performs: (1) **intent detection** across 6 types (Penalty, Definition, Procedure, Requirement, Reference, General) using Vietnamese regex patterns with LLM fallback; (2) **query type classification** into 7 categories mapped to retrieval parameters; (3) **abbreviation expansion** of Vietnamese legal abbreviations (e.g., *TNHH* $\rightarrow$ *trách nhiệm hữu hạn*); and (4) **ontology expansion** $\phi_0(q, \mathcal{O})$ via class hierarchy traversal—children, parents, and siblings of matched classes in H, with Vietnamese label lookup through L.

3.5.2. Tier 1: LLM-Guided Tree Traversal

Tier 1 navigates the document forest $\mathcal{T}$ using LLM reasoning in three loops:

1. **Loop 0** (Forest $\rightarrow$ Document): LLM selects top 3 relevant documents from document sum-

maries.

2. **Loop 1** (Document → Chapter): LLM selects top 6 relevant chapters using chapter summaries and detected keywords.

3. **Loop 2** (Chapter → Article): LLM identifies top 7 specific articles from article summaries within selected chapters.

General Provisions chapters are automatically included when specific chapters are selected. Each loop produces a reasoning path for interpretable retrieval.

3.5.3. Tier 2: Dual-Level Semantic Retrieval

Tier 2 performs hybrid semantic search over $\mathcal{G}$ and $\mathcal{O}$ with six scoring components.

Low-level (entity-specific): (1) keyphrase matching against article text, (2) cosine similarity between query and entity embeddings, (3) intent-aware PPR on the knowledge graph, and (4) ontology class matching via $\mathcal{O}$ expansion.

High-level (theme/concept): (5) legal theme identification (penalties, rights, procedures), and (6) hierarchy traversal through $\mathcal{O}$'s class taxonomy.

The combined Tier 2 score is:

$$S_{\text{T2}} = 0.05S_{\text{kp}} + 0.20S_{\text{con}} + 0.25S_{\text{PPR}} + 0.20S_{\text{sem}} + 0.15S_{\text{thm}} + 0.15S_{\text{hier}} \quad (4)$$

Multi-Intent Personalized PageRank. The PPR component computes query-specific node importance on $\mathcal{G}$:

$$\text{PPR}(v) = (1-\alpha) \sum_{u \in N(v)} \frac{w(u,v) \cdot \text{PPR}(u)}{d(u)} + \alpha \cdot p(v) \quad (5)$$

where $\alpha = 0.15$ is the teleport probability, $w(u,v)$ is the edge weight adjusted by query intent (Table 2), $d(u)$ is the weighted out-degree of u, and $p(v)$ is the personalization vector seeded from matched entities.

3.5.4. Tier 3: Semantic Bridge (RRF Fusion)

Tier 3 merges results from Tier 1 and Tier 2 using Reciprocal Rank Fusion:

$$\text{RRF}(d) = \sum_{s \in \{T1, T2, KG\}} w_s \cdot \frac{1}{k + \text{rank}_s(d)} \quad (6)$$

where $k = 60$ and source weights are $w_{T1} = 0.8$, $w_{T2} = 1.0$, $w_{KG} = 1.2$. RRF handles heterogeneous score distributions—LLM discrete selections vs. continuous semantic scores—without calibration.

Adjacent article expansion. After RRF scoring, we expand ± 2 articles from each selected article with a decay factor of 0.7. Error analysis showed failures often have correct answers within ± 3 articles of retrieved results. Top-ranked articles are then passed to an LLM for answer generation with citation instructions.

4. Experiments

4.1. Experimental Setup

Dataset. We evaluate on Vietnam Enterprise Law 2020 (No. 59/2020/QH14) and 9 related decrees: 10 chapters, 218 articles in the primary law. The evaluation set comprises 375 real questions from Vietnamese legal consultation platforms (LuatVietnam, ThuVienPhapLuat, legal forums), with expert-annotated ground truth (1–8 relevant articles per question, average 2.3). Question categories: Article lookup (41.6%), Guidance document (26.1%), Situation analysis (20.0%), Compare regulations (8.5%), Case law lookup (3.8%).

Knowledge graph statistics. Table 3 summarizes the constructed knowledge graph.

Table 3: Knowledge graph statistics.

Entity Type	Count	Relation Type	Count	Structure	Count
Total Entities	1,299	Total Relations	2,577	Document Tree Nodes	1,847
Organization	187	requires	423	Chapter Summaries	10
Person/Role	156	has_authority	312	Article Summaries	218
Legal Term	234	applies_to	287		
Legal Reference	198	references	265		
Condition	178	includes	234		
Action	163	defines_as	198		
Other types	183	Other relations	858		

Implementation. Claude 3.5 Haiku for tree traversal and query analysis; Gemini 2.0 Flash for entity extraction. Embeddings: `paraphrase-multilingual-MiniLM-L12-v2` [22]. PPR: iterative computation, max 50 iterations, tolerance 10^{-6}. All LLM responses cached in SQLite ($\sim$1.1 GB) for reproducibility.

Metrics. Hit@k, Recall@k, Precision@k,

NDCG@k at $k \in \{1, 3, 5, 10, 20, 30, 50\}$, and MRR.

4.2. Baselines

BM25 [1]: Okapi BM25 with Vietnamese tokenization ($k_1 = 1.2$, $b = 0.75$). **TF-IDF** [11]: With Vietnamese word segmentation and cosine similarity. **Semantic Search** [22]: Dense retrieval using multilingual sentence-transformers. **LightRAG** [5]: Graph-based RAG with automatic entity extraction (default parameters). **PageIndex** [14]: Our Tier 1 in isolation—LLM-guided tree traversal without semantic or KG components.

4.3. Main Results

Table 4 presents Hit@k performance.

Table 4: Hit@k performance (%). Best **bold**, 2nd underlined.

Method	@5	@10	@20	@30	@50	Avg. Art.
Semantic	7.4	15.8	38.5	44.3	48.3	50.0
BM25	15.0	26.7	50.1	58.1	79.2	50.0
TF-IDF	15.8	28.8	50.7	62.0	82.3	50.0
LightRAG	18.8	33.9	51.0	66.2	70.3	32.8
PageIndex	31.7	35.2	35.2	35.2	35.2	3.9
3-Tier (ours)	**68.5**	**75.2**	**82.1**	**84.0**	**86.4**	52.3

Key observations: (1) Semantic search performs poorly (15.8% Hit@10) due to general-purpose embeddings failing on Vietnamese legal semantics. (2) BM25/TF-IDF achieve moderate performance (∼27–29% Hit@10) but benefit from high recall at large k, indicating vocabulary overlap with weak ranking. (3) LightRAG improves over lexical methods (33.9%) through entity-relation modeling but lacks legal domain precision. (4) PageIndex achieves 35.2% Hit@10 with only 3.9 articles on average, demonstrating targeted but limited retrieval. (5) Our 3-Tier system achieves 75.2% Hit@10 (+41.3pp over the best baseline).

Figure 2 visualizes the consistent advantage across all k values.

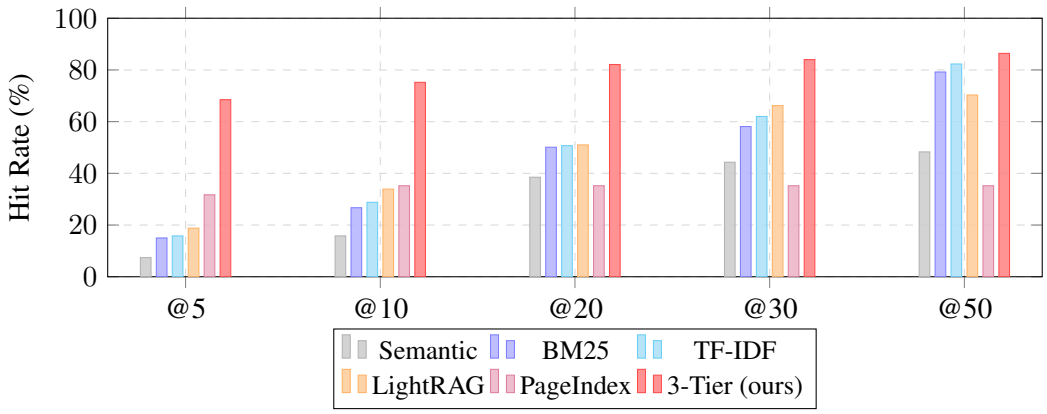

Figure 2: Hit@k comparison. Our 3-Tier system consistently outperforms all baselines, with the largest gap at Hit@5 (68.5% vs. 31.7%).

4.4. Detailed IR Metrics

Table 5 reports comprehensive retrieval metrics for our system.

Table 5: Detailed IR metrics for 3-Tier system.

k	Hit@k	Recall@k	Prec.@k	NDCG@k
1	42.7	23.9	42.7	42.7
3	62.9	41.5	26.6	41.1
5	68.5	46.7	18.3	43.4
10	75.2	51.8	10.3	45.4
20	82.1	59.2	6.1	47.8
30	84.0	61.9	4.3	48.6
50	86.4	67.4	2.9	49.9

MRR is 0.545, indicating the first relevant article appears on average at rank ∼1.8. Recall improves steadily up to $k = 50$, reaching 67.4%.

4.5. Ablation Study

Table 6 isolates each tier's contribution.

Table 6: Ablation study: per-tier contribution (%).

Configuration	Hit@10
Tier 1 only (Tree Traversal)	56.2
Tier 2 only (DualLevel + PPR)	61.7
Tier 1 + Tier 2 (no RRF fusion)	71.2
Full 3-Tier (with Tier 3 RRF)	**75.2**

Each tier provides additive improvement: Tier 1 alone achieves 56.2%, adding Tier 2 yields 71.2% (+15.0pp), and Tier 3 fusion adds 4.0pp to reach 75.2%. Among successful retrievals, 70% of hits orig-

inate from Tier 1, while Tier 2 contributes 14% of unique hits not found by Tier 1, confirming complementarity.

4.6. Analysis

4.6.1. Performance by Query Type

Table 7 breaks down Hit@10 by question category.

Table 7: Hit@10 by question category (%).

Category	Hit@10	Count
Article lookup	82.1	156
Guidance document	73.5	98
Situation analysis	71.3	75
Compare regulations	68.8	32
Case law lookup	61.1	18

The system excels on article lookup queries (82.1%) where Tier 1's structured navigation directly identifies targets. Complex categories like situation analysis and compare regulations benefit from Tier 2's multi-hop reasoning through the knowledge graph.

4.6.2. Qualitative Analysis

Figure 3 illustrates the three-tier process on a representative query.

Success case. Query: "Conditions for establishing a joint-stock company?" Tier 1 navigates Chapter II → Articles 111–112, correctly identifying JSC establishment conditions. Tier 2 finds additional Article 17 (general conditions) via *requires* in KG. Tier 3 fuses both, ranking Articles 111, 112, 17 in top 3. Ground truth: Articles 111, 112—**Hit@3**.

Failure case. Query: "Penalty for late annual report submission?" Tier 1 identifies Enterprise Law articles on reporting obligations but misses the penalty, which resides in Decree 122/2021, a separate document. Tier 2's KG expansion partially reaches the decree via *has_penalty* edges but ranks the penalty article at position 15. This illustrates the challenge of cross-document retrieval when answers span multiple legal instruments.

4.6.3. Error Analysis

Analysis of the 93 failed queries (24.8% of total) reveals three main failure modes: (1) **Cross-document gaps** (42% of failures): the answer resides in a decree not directly linked to the primary law, requiring deeper cross-reference chains. (2) **Vietnamese compound term splitting** (28%): word segmentation errors fragment compound legal terms (e.g., "trách nhiệm hữu hạn" split incorrectly), degrading entity matching. (3) **Ambiguous intent** (30%): queries mixing multiple intents (e.g., "conditions *and* penalties for...") cause intent-weighted PPR to prioritize one path over the other.

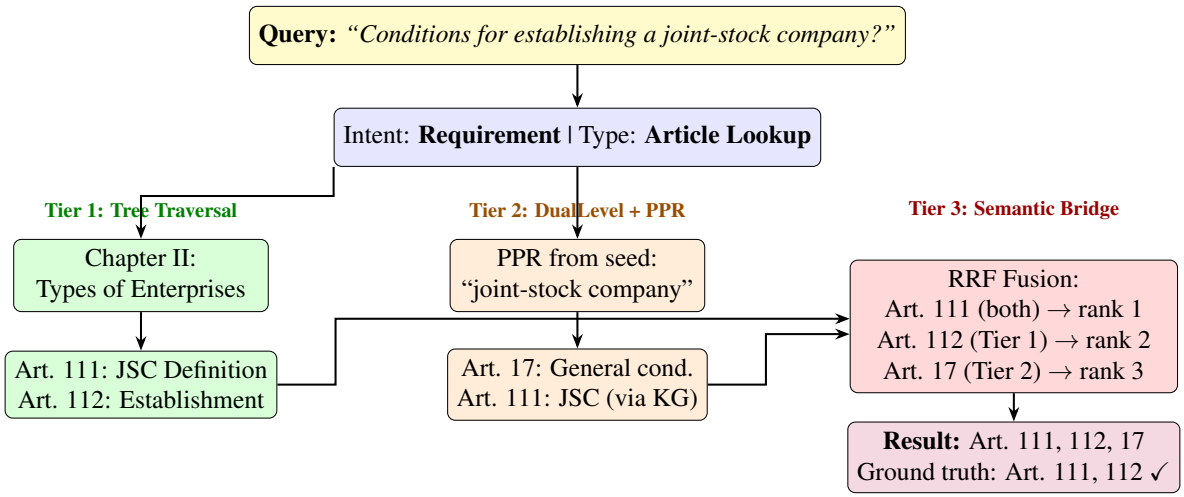

Figure 3: Query walkthrough. Tier 1 navigates the document tree, Tier 2 uses PPR on the knowledge graph, and Tier 3 fuses results via RRF.

Reproducibility. All LLM responses are cached in SQLite (∼1.1 GB), making results fully deterministic. Standard deviations are zero and not reported.

5. Discussion

Three findings emerge from our experiments. First, the tiers are **complementary rather than redundant**: Tier 1 excels at focused article identification via structural reasoning (70% of hits), while Tier 2 captures cross-reference and thematic relationships (14% unique hits). The 15pp gain from adding Tier 2 to Tier 1 confirms that structural and semantic retrieval address different failure modes.

Second, **intent-aware PPR** is particularly effective for situation analysis and guidance queries, where traversing the correct relation types (e.g., *requires* for

procedures, *has_authority* for rights) substantially improves ranking. The weight matrix (Table 2) encodes legal domain knowledge that general-purpose graph traversal lacks.

Third, **document structure matters more than flat search**: PageIndex (35.2% Hit@10 with 3.9 articles) matches LightRAG (33.9% with 32.8 articles), retrieving equivalent accuracy with an order of magnitude fewer articles. This validates the value of hierarchical navigation in legal corpora.

The knowledge representation model $\mathcal{M}$ provides a principled framework for organizing legal knowledge across complementary representations. By separating terminological ($\mathcal{O}$), assertional ($\mathcal{G}$), and structural ($\mathcal{T}$) knowledge, each retrieval tier can specialize in its strength while the fusion tier integrates their contributions.

6. Conclusion

We presented a three-tier knowledge graph RAG system for Vietnamese legal QA that extends the Legal-Onto Rela-model with a formal knowledge model $\mathcal{M} = (\mathcal{O}, \mathcal{G}, \mathcal{T}, \Phi)$. The system combines LLM-guided structural navigation, dual-level semantic retrieval with intent-aware Personalized PageRank, and Reciprocal Rank Fusion. Experiments on 375 real questions demonstrate 75.2% Hit@10 (MRR 0.545), outperforming baselines by 41.3 percentage points. Ablation studies confirm each tier's additive contribution.

Limitations. Tree traversal requires multiple LLM calls per query ($\sim$7s latency). Entity extraction accuracy for Vietnamese legal text has not been independently validated. Results are specific to Enterprise Law; other legal domains may exhibit different retrieval patterns.

Future work. We plan to: (1) extend the knowledge graph to cover related laws (Investment, Tax, Labor) with cross-law entity resolution, (2) add multi-turn conversation support for follow-up questions, (3) develop advanced 3+ hop KG reasoning for complex transitive inference, and (4) build explainable retrieval traces showing tier and ontology path contributions.

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